

Personal Firewall using Python Report

1. Introduction

A personal firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. It helps protect a computer from unauthorized access, malware, and suspicious network activity. This project focuses on developing a lightweight personal firewall using Python that can filter network traffic based on IP address, port number, and protocol type.

2. Abstract

This project presents a Personal Firewall developed using Python and Scapy to monitor and filter network traffic. The firewall captures incoming and outgoing packets and checks them against predefined rules to allow or block traffic. Suspicious packets are logged for security analysis. The system can also integrate with iptables in Linux to enforce firewall rules at the system level. An optional graphical interface can be created using Tkinter for live traffic monitoring and rule management. This project helps in understanding network security and packet filtering techniques.

3. Tools Used

Python, Scapy, iptables (Linux), Tkinter (for GUI), Networking Tools, Log Files.

4. Steps Involved in Building the Project

1. Install Python and Scapy library. 2. Capture network packets using Scapy. 3. Define firewall rules to allow or block packets based on IP, port, and protocol. 4. Analyze each packet and compare it with rule sets. 5. Allow safe packets and block suspicious packets. 6. Log blocked packets with details such as IP address, port, and protocol. 7. Optionally, integrate iptables to enforce rules at the system level. 8. Create a Tkinter GUI for live monitoring and rule customization. 9. Test the firewall with different traffic scenarios.

5. Conclusion

The Personal Firewall using Python successfully monitors and filters network traffic based on predefined rules. It helps in blocking suspicious traffic and logging network activity for security analysis. This project improves understanding of firewall systems, packet filtering, and network security. The system can be further enhanced by adding automatic rule generation, alert notifications, and advanced traffic analysis features.